Predicting Short-Term Popularity Growth of YouTube Trending Videos Using Platform Metrics and External Search Interest Signals

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Abstract

This project investigates whether external search interest data from Google Trends can improve the prediction of short-term popularity growth of YouTube trending videos. Using publicly available YouTube Trending data enriched with category-level Google Trends signals, the study applies exploratory data analysis, statistical hypothesis testing, and supervised machine learning models. Results indicate that while overall predictive performance remains moderate, incorporating Google Trends features provides a consistent improvement, particularly in precision–recall performance for identifying high-growth videos.

1. Introduction and Motivation

YouTube is one of the largest online media platforms, where videos can experience rapid and sometimes unpredictable popularity growth. Understanding the factors that drive short-term surges in attention is valuable for content creators, advertisers, and analysts. While platform-level engagement metrics (e.g., views, likes, comments) provide strong signals, they may not fully capture broader public interest outside the platform. This project explores whether external search behavior, measured via Google Trends, can complement platform metrics in predicting short-term popularity growth among already-trending videos.

2. Research Questions

RQ1: Which video- and channel-level features are associated with higher short-term popularity growth?
RQ2: Does Google Trends search interest improve predictive performance?
RQ3: Can supervised machine learning models reliably classify high-growth videos?

3. Data Sources

3.1 YouTube Trending Dataset

The primary dataset is the YouTube Trending Videos dataset obtained from Kaggle (US subset). It includes video metadata and engagement metrics such as views, likes, dislikes, comment counts, publication time, and category identifiers.

3.2 Google Trends Data

Google Trends data was collected programmatically using the pytrends Python library. Representative search topics were defined per YouTube category, and daily search interest scores (0–100) were retrieved. Category-level time series were merged with the YouTube dataset based on each video’s trending date. All Google Trends features were derived specifically for this project (not downloaded as a pre-existing dataset).

4. Exploratory Data Analysis

Exploratory analysis shows that engagement variables (views, likes, dislikes, comment_count) are highly skewed, with a small number of videos receiving extremely high engagement. Category-level counts indicate that some categories dominate the trending feed. To reduce scale effects, engagement ratios such as like-to-view and comment-to-view were used in later stages. As shown in Figure 3, engagement ratios exhibit right-skewed distributions with relatively stable central tendencies, supporting their use as normalized predictors in subsequent modeling stages. Due to this heavy right-skew, engagement distributions were visualized on a log10 scale to better capture relative differences across the majority of videos.

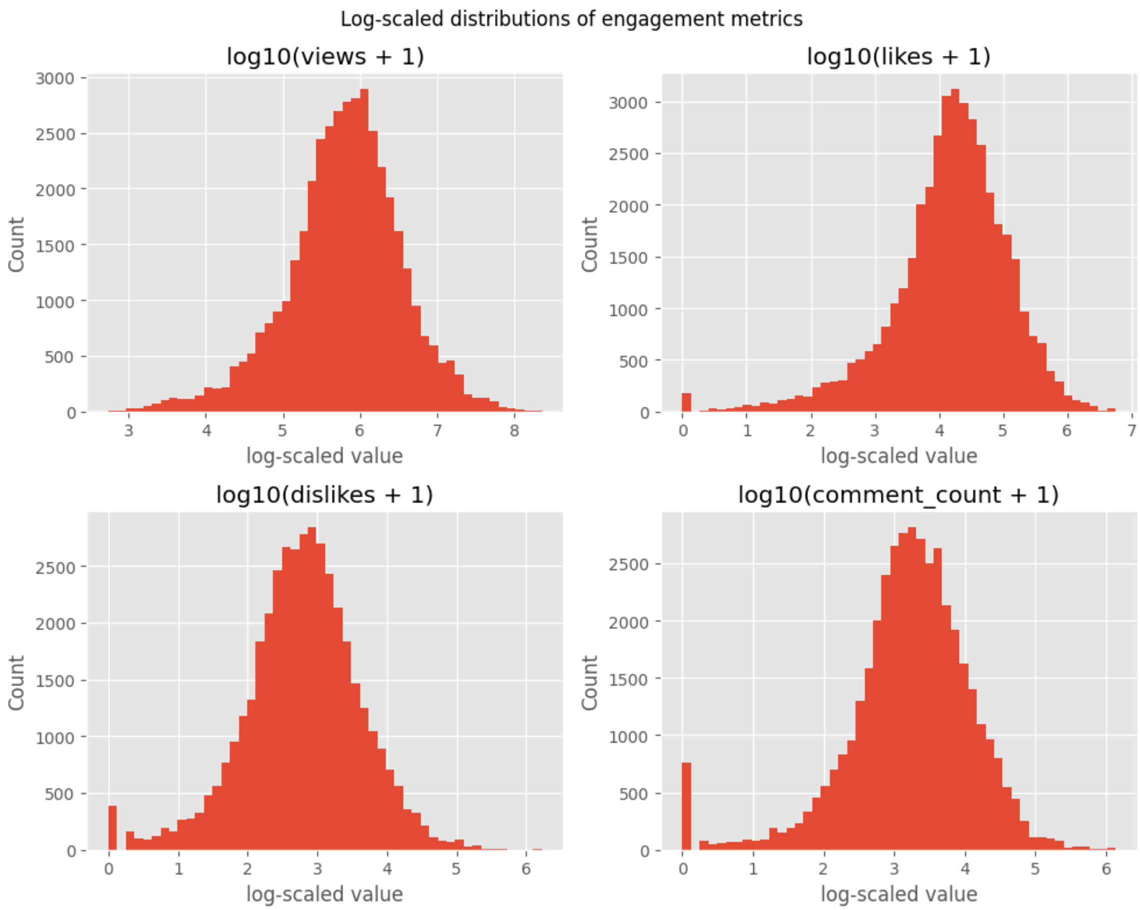

Figure 1. Distributions of engagement metrics in the US trending dataset (from 01_eda.ipynb).

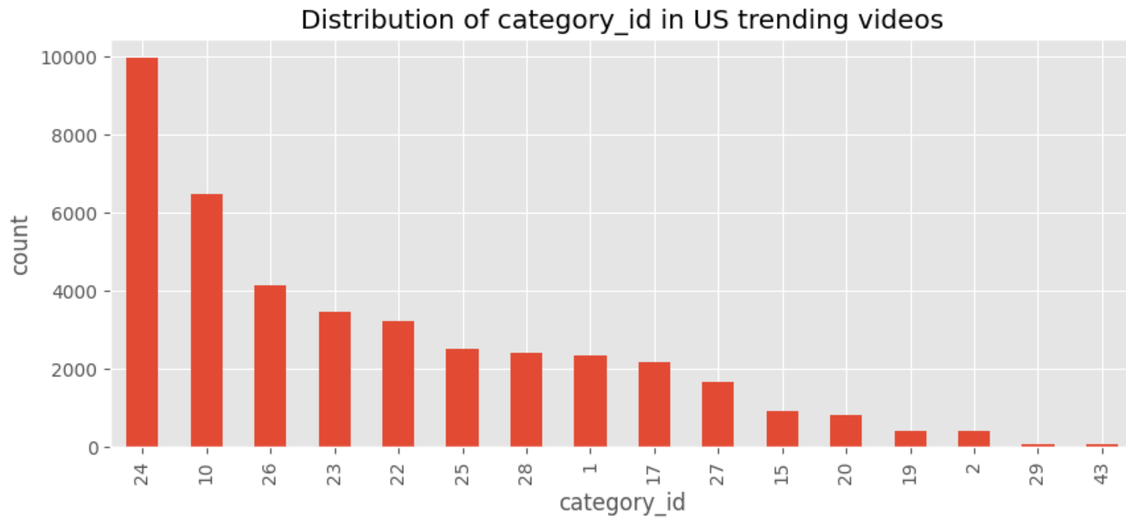

Figure 2. Category distribution for US trending videos (from 01_eda.ipynb).

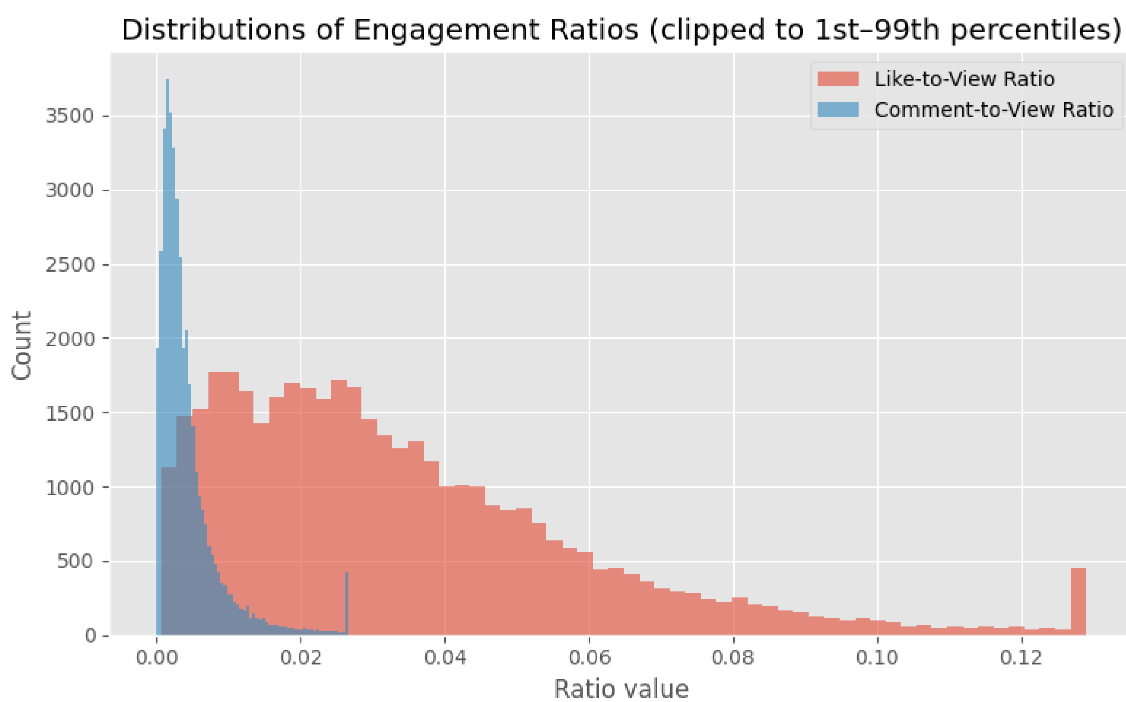

Figure 3. Distributions of engagement ratios (from 01_eda.ipynb).

5. Feature Engineering

Features were engineered from raw engagement and metadata. Core predictors include views, likes, dislikes, comment counts, publication hour, and engagement ratios (like/view and comment/view). Google Trends features include category-level daily trend scores and rolling averages (3-day and 7-day means). A binary target variable, high_growth, was constructed using category-normalized next-day view growth. Growth-based variables were used only to define the target and were excluded from the model feature set to avoid target leakage.

6. Statistical Analysis

To test whether Google Trends search interest differs between high-growth and low-growth videos, a non-parametric Mann–Whitney U test was conducted on Google Trends scores. The test indicates a statistically significant difference ($p < 0.001$), with higher average trend scores observed for the high-growth group.

In addition to Google Trends search interest, non-parametric hypothesis tests were also applied to engagement ratios (like-to-view) and publication timing (publish hour). A Mann–Whitney U test indicates that high-growth videos exhibit significantly higher like-to-view ratios, while a Kolmogorov–Smirnov test reveals a significant difference in the distribution of publish hours between high-growth and low-growth videos.

Table 1. Mann–Whitney U test summary (Google Trends score vs. high_growth).

Rows used	Mann–Whitney U	p-value	Mean trend (high vs low)
33,409	111,591,307.50	< 0.001	66.683 vs 63.701

7. Machine Learning Models

Supervised learning models were trained to classify high-growth videos: Logistic Regression, Random Forest, XGBoost, and LightGBM. A leakage-aware split was used at the video level to prevent the same video from appearing in both training and test sets. Performance was evaluated using accuracy, F1-score, ROC-AUC, and PR-AUC, with emphasis on PR-AUC due to class imbalance. Because the high-growth class represents a minority of observations, precision–recall–based metrics are emphasized to better reflect model performance under class imbalance. Although ensemble models are generally expected to outperform linear models on non-linear datasets, logistic regression achieved comparable or slightly better performance in this study. This can be attributed to the relatively low feature dimensionality, the dominance of monotonic and ratio-based predictors, and the presence of noise and class imbalance, which may reduce the advantages of more complex models. Figure 4 illustrates the precision–recall curves of the evaluated models, showing that models incorporating Google Trends features achieve improved precision at higher recall levels, which is particularly important given the class imbalance in high-growth prediction.

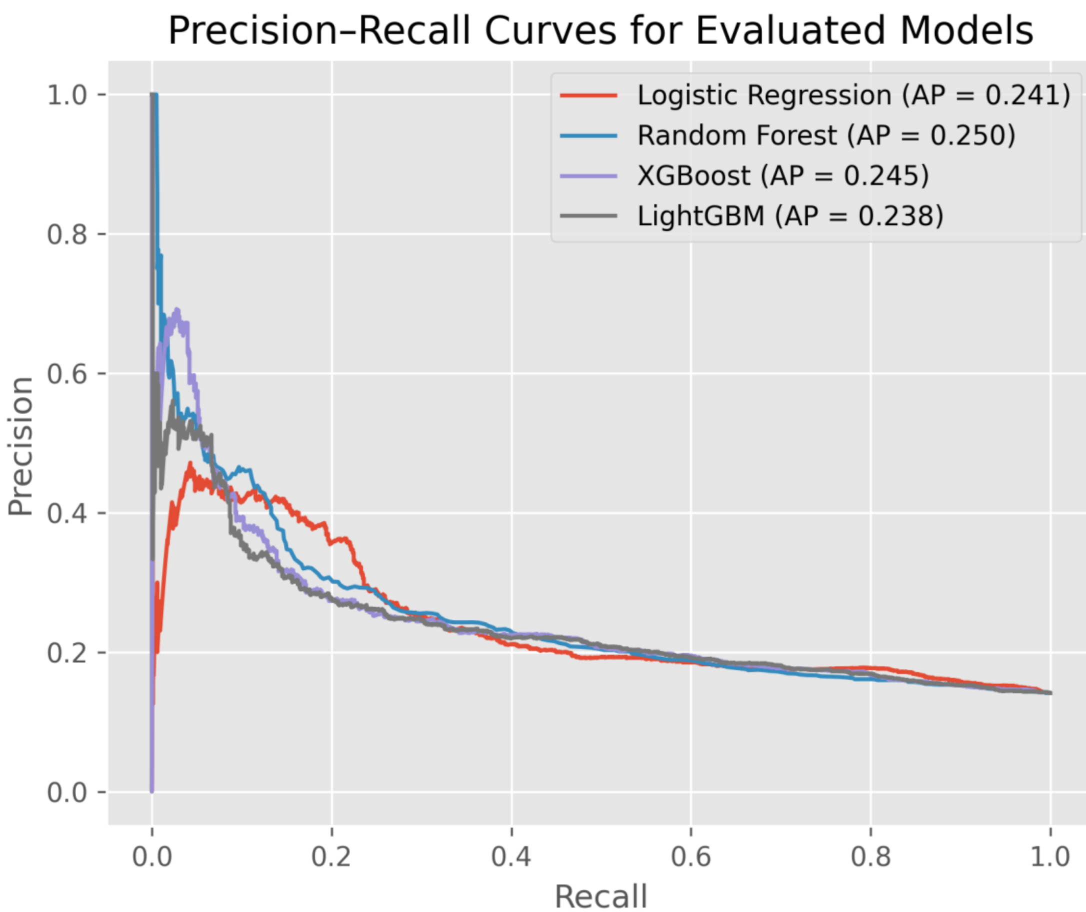

Figure 4. Precision–recall curves for the evaluated machine learning models. Models incorporating Google Trends features achieve higher precision at comparable recall levels, consistent with observed improvements in PR-AUC under class imbalance (from 03_modeling.ipynb).

8. Results

Two feature sets were evaluated: (A) YouTube platform metrics only, and (B) YouTube metrics plus Google Trends features. Incorporating Google Trends provides a modest but consistent improvement in predictive performance, particularly in PR-AUC. Although the absolute gains are modest, the consistent improvement across metrics suggests that Google Trends features add complementary information beyond platform engagement signals.

Table 2. Test-set performance for feature sets (from 03_modeling.ipynb).

Feature Set	ROC-AUC	PR-AUC
YouTube only	0.6139	0.3353
YouTube + Trends	0.6197	0.3662

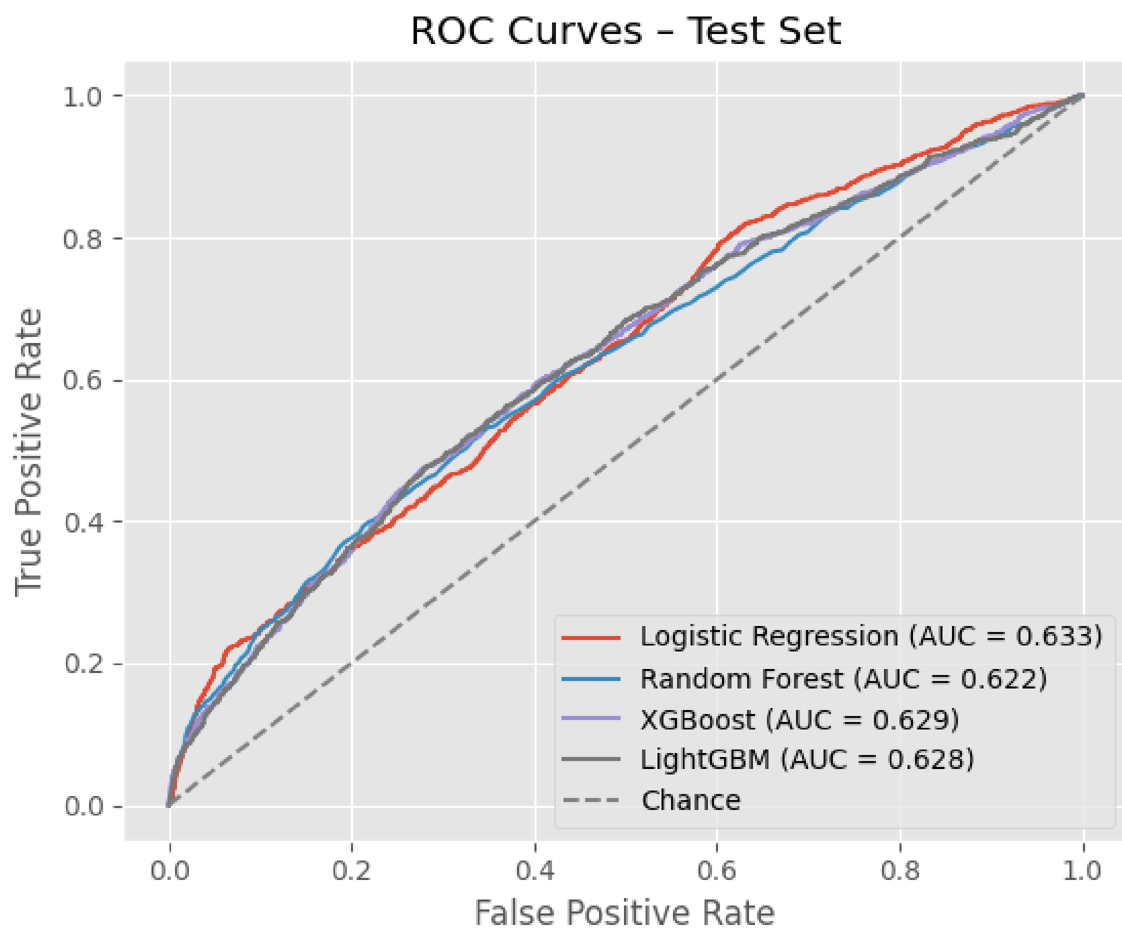

Figure 5. ROC curve figure (from 03_modeling.ipynb).

9. Model Interpretation

Feature importance analysis for tree-based models suggests that engagement ratios and trend signals can contribute to decision-making, though the overall effect of Google Trends features is smaller than core platform engagement predictors. The observed improvement is more pronounced in PR-AUC, consistent with the goal of identifying a minority high-growth class. For logistic regression, coefficient magnitudes were used as an interpretable measure of feature importance. This analysis highlights the role of category, publication timing, and trend-related features, demonstrating that linear models provide both competitive performance and transparent interpretability.

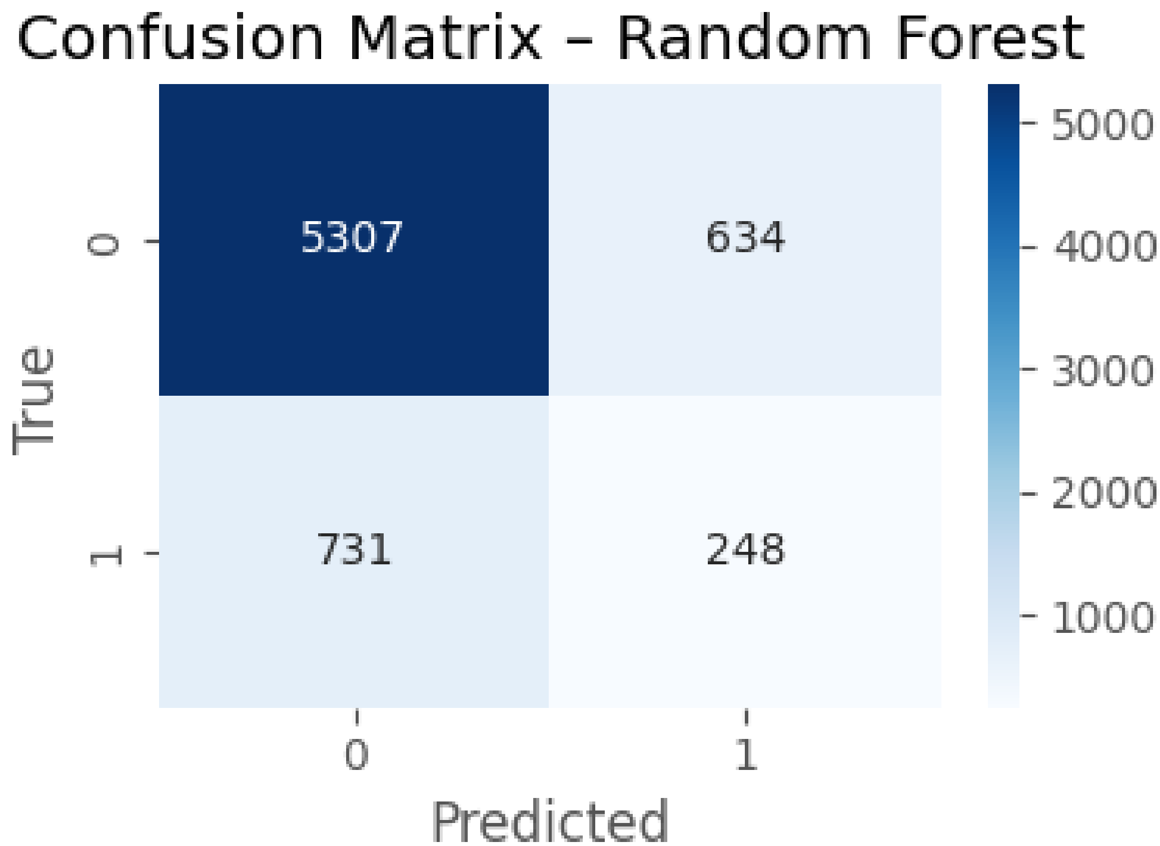

Figure 6. Feature importance visualization from 03_modeling.ipynb output.

10. Limitations and Future Work

This study is limited by the nature of the YouTube Trending dataset, which includes only already-trending (curated) content and may not represent the full YouTube ecosystem. Google Trends measures general topic interest rather than interest in individual videos. Future work could incorporate NLP features from titles/descriptions/tags, multi-country datasets, and time-series models to better capture temporal dynamics and exogenous signals.

11. Ethical Considerations

All data used in this project is publicly available and aggregated. No personal or private user data was collected. Potential biases may arise from YouTube's trending mechanism, which can favor large channels or specific content categories. Findings should therefore be interpreted as patterns within a curated subset of content.

12. Conclusion

External search interest signals captured via Google Trends provide complementary information for predicting short-term popularity growth among trending YouTube videos. While overall predictive performance remains moderate, the consistent improvement—especially in PR-AUC—suggests that exogenous attention signals can modestly enhance platform-based models.

13. AI Usage Disclosure

AI tools were used in a limited and transparent manner during this project. Specifically, AI assistance was utilized for drafting and refining documentation, brainstorming ideas, and debugging code logic. All data processing, feature engineering, modeling decisions, and result interpretations were performed by the author. No AI-generated outputs were used directly as final results without verification. Details of AI usage, including example prompts and generated outputs, are documented in a separate ai_usage.md file.

14. References

Kaggle – YouTube Trending Videos dataset (US subset).
https://www.kaggle.com/datasets/datasnaek/youtube-new
Google Trends via pytrends Python library.